Longitudinal Patient Health Prediction - from previous visits to the next visit

Question: *given a patient's history up to the current visit, what outcome is likely at the next visit, and how reliable is that prediction?* | Academic research prototype - not a clinically validated tool.

1 Data (public, real)

- **GSE65391** - paediatric SLE cohort (NCBI GEO)
- 157 patients, 923 visit records (avg 5.9 per patient)
- Per visit: 14 labs, 8 treatment flags, SLEDAI + 6 organ items, age
- Usable prediction points: **635** (111 patients with ≥3 records); flares = **17.2 %**
- No calendar dates: time = days since the patient's first record



2 Target (one clear outcome)

$$y_{t+1} = 1 \text{ if } SLEDAI_{t+1} - SLEDAI_t \geq 4 \text{ else } 0$$

- "Flare at the next record" (≥4-point rise is a common SLEDAI flare cut-off)
- A proxy label derived from SLEDAI - **not** clinician-adjudicated
- Needs ≥2 records (current + one earlier) to measure change
- Not predicted: readmission, complications, abnormal labs (not supported by this dataset)

3 Leakage-free features (87)

$$\Delta x_t = x_t - x_{t-1} \quad slope = \Delta x_t / gap \times 30 \quad roll3 = mean(x_{t-2..t})$$

- Trends for 11 variables (previous value, change, slope, 3-visit mean)
- History: highest/mean SLEDAI so far, flares seen so far, visits since last flare, treatment changes
- **Excluded as future information:** total visit count, span to last visit, biopsy timing/class, study split
- **Unit tests prove a visit's features do not change if later visits are altered**

4 Models & validation

Model	Role
Logistic Regression	baseline (L2, C=0.1)
Random Forest	300 trees, leaf ≥5
XGBoost	depth 3, 200 trees
Average-rate / SLEDAI-only	sanity baselines
LSTM / GRU	not used - too few patients

79 train patients - grouped 5-fold CV x3 | 32 held-out patients

split by PATIENT - no patient appears on both sides; forward-in-time check inside each patient

5 Metrics

$$Precision = TP/(TP+FP) \quad Recall = TP/(TP+FN) \quad F1 = 2PR/(P+R)$$

$$Brier = mean\,(p - y)^2 \quad ECE = \Sigma\,(n_b/n)\,|obs_b - pred_b|$$

- AUROC and PR-AUC (chance PR-AUC = flare rate 0.17)
- Threshold chosen on out-of-fold training data only
- Platt scaling for calibration; 95 % CIs by resampling patients (1000x)

6 Uncertainty & explanation

$$f(x) = \varphi_0 + \Sigma_i\,\varphi_i \;(SHAP) \quad \hat{p} = mean_{b \in OOB}\,p_b$$

- 60 bootstrap-by-patient models; a known patient is scored only by models that never saw them
- Spread of the models (10-90 %) = reliability band
- SHAP groups: recent trend / longer history / treatment / current values

Association ≠ causation: SHAP shows what the model uses, not what would happen if a treatment changed.

7 Results snapshot (honest)

Evaluation	AUROC	PR-AUC
Held-out patients, longitudinal LR	0.65 (0.49-0.77)	0.33
Out-of-bag, all patients (deployed)	0.63 (0.58-0.68)	0.27
Static vs longitudinal (CV, LR)	0.67 vs 0.64	0.30 vs 0.26
Chance	0.50	0.17

- Signal is **modest**; history features did **not** clearly beat current-visit features
- Calibration after Platt: ECE 0.09 → 0.03 (held-out)

8 Contribution (claimed carefully)

- Leak-audited, patient-split next-visit pipeline on a public cohort, with tests
- Direct static-vs-longitudinal comparison with paired patient-level intervals
- Per-patient out-of-bag prediction, reliability band and trend-level explanation in one app
- **Not claimed:** first SLE flare model, or that history improves accuracy (not shown here)

9 App (Streamlit)



- <2 records → clear "insufficient history" message
- Future records are never shown as known (hindsight only behind a labelled toggle)

Workflow

Raw GEO file GSE65391 series matrix → **Clean visits** SLE only, drop replicates → **Sort in time** by patient ID + time → **Features(t)** visits ≤ t only → **Target(t+1)** SLEDAI rise ≥ 4 → **Split by patient** train / held-out → **Compare models** static vs longitudinal → **Calibrate** Platt + threshold → **Bootstrap ensemble** 60 models, OOB → **Streamlit app** risk + SHAP + limits